

CHAPTER

55



Syphilis and Gonorrhoea

Learning Objectives

- ⇒ To learn about syphilis and gonorrhea
- ⇒ To know about the causative factors of above said diseases
- ⇒ To learn about the clinical features of the above said diseases
- ⇒ To learn about the differential diagnosis of above said diseases
- ⇒ To know about the necessary physical examination performed in the patients
- ⇒ To learn about the laboratory investigations performed in the patients and interpreting the results of the laboratory investigations

Introduction

Syphilis and gonorrhea are considered as sexually transmitted diseases (STD) as these diseases are transmitted to the non infected individual by the sexual contact of the infected individual. Syphilis is caused by spirochete *Treponema pallidum*. Gonorrhea is caused by bacteria *Neisseria gonorrhoeae*. Each of the infections are described separately.

Syphilis is characterized by 3 sequential clinical, symptomatic stages separated by periods of asymptomatic latent infection. Common manifestations include genital ulcers, skin lesions, meningitis, aortic disease and neurologic syndromes. Diagnosis is done by serologic tests and adjunctive tests selected based on the stage of the disease.

Gonorrhea infects the epithelia of urethra, cervix, rectum, pharynx or eyes causing irritation or pain and purulent discharge from the urethra or vagina. Dissemination to skin and joints, which is uncommon, causes sores on the skin, fever and migratory polyarthrititis or pauciarticular skin, fever and migratory polyarthrititis or pauciarticular septic arthritis. Diagnosis is done by microscopy, culture or nucleic acid amplification tests.

Now we are going to learn each of the infections in a little bit more detail.

Syphilis

As it is being mentioned that this sexually transmitted

infection is caused by *Treponema pallidum*, a spirochete that cannot survive outside the human body for long time. This microbe enters through mucous membranes of the skin, reaches the regional lymph nodes within hours and rapidly spread throughout the body.

- ⊗ It is usually transmitted by sexual contact (genital, orogenital and anogenital), but may be transmitted nonsexually through skin contact or transplacentally. Risk of transmission is about 30% from a single sexual encounter with a person who has primary syphilis and 60 to 80% from an infected mother to a foetus.

Stages of syphilis

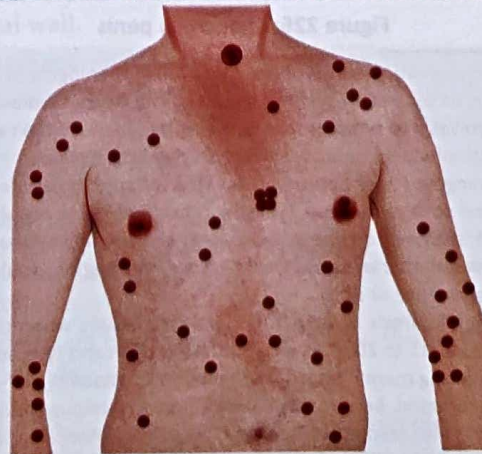


Figure 225: Features of syphilis

Syphilis occurs in primary, secondary and tertiary stages with long latent periods between them. Infected individuals are contagious during first two stages.

Clinical features

Syphilis (see Figure 225) may manifest at any stage and may affect multiple or single organs, mimicking many other disorders. Syphilis may be accelerated by coexisting HIV infection, wherein eye involvement, meningitis and other neurologic complications are more common and more severe.

Primary syphilis

- After an incubation period of 3 to 4 weeks, a primary lesion (chancre) develops at the site of inoculation. The initial red papule quickly forms a chancre, usually a painless ulcer with a firm base; when rubbed, it produces a clear fluid containing numerous spirochetes. Chancres can occur in any part of the body, but are more common among the following:
 - Penis, anus and rectum in men
 - Vulva, cervix, rectum and perineum in women
 - Lips or mouth in either gender



Figure 226: Chancre in penis

Secondary syphilis

- The spirochetes spread in the blood stream, producing widespread mucocutaneous lesions, lymph node swelling and symptoms of other organs. Symptoms typically begin 6 to 12 weeks after the appearance of chancre. Fever, loss of appetite, nausea, and fatigue are common. Headache, hearing or balancing disturbances and pain in bones also occur.

Tertiary or late syphilis

- About 1/3rd of untreated people develop late syphilis, although not until years to decades after the initial infection. Lesions may be clinically classified as benign tertiary syphilis, cardiovascular syphilis or neurosyphilis.
- **Benign tertiary gummatous syphilis:** usually develops within 3 to 10 years of infection, may involve the skin, bones and internal organs. Gummas are soft, destructive, inflammatory masses that are typically localized but may diffusely infiltrate an organ or tissue; they grow and heal slowly and leave scars.
- **Benign tertiary syphilis of bone:** results in either inflammation or destructive lesions that cause a deep, boring pain, characteristically worse at night.
- **Cardiovascular syphilis:** usually manifests 10 to 25 years after the initial infection as aneurysmal dilatation of ascending aorta, insufficiency of aortic valve, or narrowing of coronary arteries. Pulsations of dilated aorta may cause symptoms by compressing or eroding adjacent structures in the chest. Symptoms produced may be brassy cough, infections and breathing discomfort secondary to compression of trachea, hoarseness of voice due to vocal cord paralysis and painful erosion of sternum and spine.
- **Neurosyphilis:** This can manifest in several forms:
 1. Asymptomatic neurosyphilis¹
 2. Meningovascular neurosyphilis²
 3. Parenchymatous neurosyphilis³
 4. Tabes dorsalis⁴

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- 1 Asymptomatic neurosyphilis: causes mild meningitis in about 15% of the cases originally diagnosed as latent syphilis; in 25 to 40% of the cases as secondary syphilis; in 12% of the cases, cardiovascular syphilis and in 5% of the cases, benign tertiary syphilis. Without treatment, it evolves into symptomatic neurosyphilis in 5% of the cases. If CSF examination does not detect the evidence of meningitis 2 years after the initial infection, neurosyphilis is less likely to develop.
- 2 Meningovascular neurosyphilis: This variant of the infection results from inflammation of large to medium sized arteries of the brain or spinal cord; symptoms typically occur 5 to 10 years after infection and range from none to stroke. Initial symptoms include headache, neck stiffness, dizziness, behavioral abnormalities, poor concentration, memory loss, lassitude, insomnia and blurred vision. Spinal cord involvement may cause weakness and wasting of shoulder girdle and arm muscles, slowly progressive leg weakness with urinary or faecal incontinence or both.
- 3 General paresis or dementia paralytica results when chronic meningo encephalitis causes destruction of cortical parenchyma. It usually develops 15 to 20 years after initial infection and typically does not affect individuals before 40s or 50s. Behavior progressively deteriorates, mimicking mental disorder or dementia. Irritability, difficulty in concentration, memory disturbances, headache, insomnia, fatigue, lethargy are common. Seizures, aphasia or transient hemiparesis are possible.
- 4 This is also termed as locomotor ataxia, involves slow, progressive degeneration of the posterior columns and nerve roots. It typically develops 20 to 30 years after initial infection. The earliest characteristic symptom is an intense, stabbing pain in the back and legs that recurs irregularly. Gait ataxia, hyperesthesia and paraesthesia may produce sensation of walking on foam rubber. Urine retention, urine incontinence, recurrent infections and erectile dysfunction may be associated along with this disease.

Diagnosis is done by VDRL (Venereal Disease Research Laboratory) test.

Syphilis is suspected in patients with mucocutaneous lesions or unexpected neurologic disorders.

Gonorrhoea

As stated earlier, gonorrhea is mainly caused by *Neisseria gonorrhoeae* and is almost always transmitted by sexual contact.

Urethral and cervical infections are common, but infection in pharynx or rectum occur after oral or anal intercourse, and conjunctivitis may follow after contamination of the eye.

After an episode of vaginal intercourse, likelihood of transmission from women to men is about 20%, but from men to women, it may be higher.

Neonates can acquire conjunctival infection during the passage through birth canal, and children may acquire the infection through sexual abuse.

In about 10-20% of the women, cervical infection may ascend via endometrium to the fallopian tubes and pelvic peritoneum leading to the manifestation of pelvic inflammatory disease (PID).

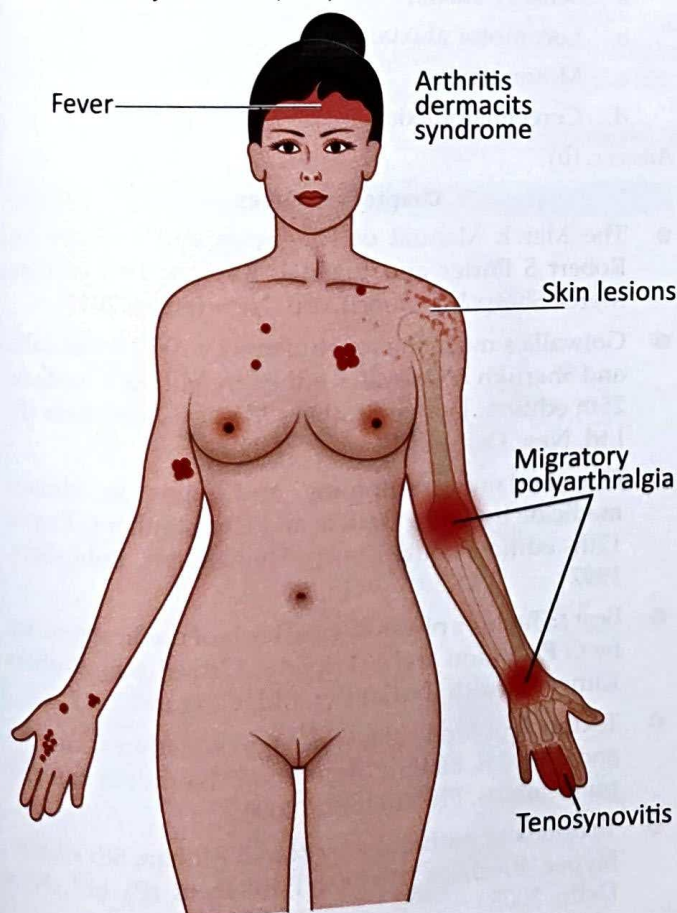


Figure 227: Arthritis dermatitis syndrome

⊕ **About 10-20%** of the infected women and very few infected men are asymptomatic. About 25% of the men will have minimum symptoms. Following are the symptoms observed in the patients suffering from gonorrhea.

⊕ **Male urethritis:** has an incubation period of 2-14 days. Onset is usually marked by mild discomfort in the urethra, followed by more severe penile tenderness and pain, dysuria and a purulent discharge.

Urinary frequency and urgency may develop as the infection spreads to the posterior urethra. Examination detects a purulent, yellow-green urethral discharge, and the meatus may be inflamed.

⊕ **Epididymitis:** usually causes unilateral scrotal pain, tenderness and swelling.

⊕ **Cervicitis:** usually has an incubation period of > 10 days. Symptoms range from mild to severe and include dysuria and vaginal discharge.

During pelvic examination, clinician may notice a mucopurulent or purulent cervical discharge, and cervical os may be red and bleed easily when touched with the speculum.

Urethritis may occur concurrently; pus may be expressed from the urethra when the symphysis pubis is pressed.

⊕ **Pelvic inflammatory disease (PID):** occurs in 10-20% of infected women. PID may include salpingitis (inflammation of fallopian tubes), pelvic peritonitis, and pelvic abscesses, may lead to lower abdominal tenderness, dyspareunia (painful intercourse), marked tenderness on palpation of abdomen, adnexae or cervix.

⊕ **Rectal gonorrhea:** is usually asymptomatic. It occurs predominantly in men practicing receptive anal intercourse and it can occur in women who participate in anal sexual intercourse. Symptoms include rectal itching, a cloudy rectal discharge, bleeding and constipation. Examination with a proctoscope may detect erythema or mucopurulent exudate on the rectal wall.

⊕ **Gonococcal pharyngitis:** is usually asymptomatic but may cause sore throat. This occurs as a result of couple involving oral sex (either phellatio or cunnilingus) with infected individual.

Diagnosis

⊕ **Gram staining and culture⁵** of genital fluids, blood or joint fluids

⊕ **Nucleic acid based testing⁶** of genital fluids, blood or joint fluids

⁵ Gram stain is sensitive and specific for gonorrhea in men with urethral discharge; gram negative intracellular diplococci are seen. Culture is sensitive and specific, but because gonococci are fragile and fastidious, sample

⁶ Nucleic acid based testing may be done on genital, rectal or oral swabs.

Key summary of the chapter

- ✱ Syphilis and gonorrhea are categorized under sexually transmitted diseases (STD) as these are transmitted through sexual contact.
- ✱ *Treponema pallidum*, a kind of spirochete responsible for the transmission of syphilis and gonorrhea is transmitted by bacteria, *Neisseria gonorrhoeae*.
- ✱ Apart from the symptoms of reproductive system, these diseases are responsible for other systemic symptoms involving nervous system, integumentary system, urinary system.
- ✱ VDRL test is performed to confirm the diagnosis of syphilis; gram staining, culture and nucleic acid based tests are performed to confirm gonorrhea.

MCQs

1. VDRL test is done to confirm the diagnosis of

- a. Gonorrhea; b. Syphilis;
- c. Dermatitis; d. AIDS

Answer: (b)

2. Syphilis is often called as

- a. The invisible dictator;
- b. The terrorizing emperor;
- c. The great imitator;
- d. The great terminator

Answer: (c)

3. *Treponema pallidum*, a causative organism of syphilis is

- a. Bacteria; b. Virus;
- c. Fungus; d. Spirochete

Answer: (d)

4. *Tabes dorsalis*, a clinical condition manifested in the patient suffering from

- a. Syphilis;
- b. Gonorrhea;
- c. Both a and b;
- d. None of the above

Answer: (a)

5. Chancre of syphilis is found in

- a. Throat, chest area and vagina;
- b. Genitals only;
- c. Buttocks, back, arms and legs;
- d. Genitals, anus, rectum, vagina, lips and mouth

Answer: (d)

6. Gonorrhea is caused in the human beings is a type of

- a. Virus; b. Bacteria;
- c. Fungus; d. Actinomycetes

Answer: (b)

7. Primary, secondary and tertiary are the types of

- a. Gonorrhea; b. Filariasis;
- c. Syphilis; d. *Pemphigus vulgaris*

Answer: (c)

8. Incubation period of primary syphilis is

- a. 1 to 6 weeks; b. 3 to 4 weeks;
- c. 6 to 8 weeks; d. 8 to 10 weeks

Answer: (b)

9. Chancre is a type of skin lesion seen in which of the following conditions?

- a. Syphilis;
- b. Filariasis;
- c. *Pemphigus vulgaris*;
- d. Scarlet fever

Answer: (a)

10. *Tabes dorsalis* is also termed as

- a. Sensory ataxia;
- b. Locomotor ataxia;
- c. Motor ataxia;
- d. Cerebellar ataxia

Answer: (b)

Chapter resources

- ✱ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ✱ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ✱ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ✱ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
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